

Geometrical shapes are found every where around us. Geometry is one of the oldest branch of mathematics which links mathematics with practical objects of our lives.

10.1 LINE SEGMENTS

Segment means a part. Line segment is a part of line. It has two distinct end points.

In the following figure we can see a line ℓ with points P and Q marked on it.



The part from P to Q of the above line ℓ is called the line segment PQ (denoted by $\overline{PQ}$ or $\overline{QP}$) and points P and Q are its end points.

How many Points are there on a line?



Line segment can be observed in our environment as edges of door, window, desk, board, table, copy, ruler etc.

Let us revise the differences between line, line segment and Ray with the help of following chart.

With respect to	Line	Line Segment	Ray
End points	No end point	Two end points	One end point
Length	No definite length	Definite length	No definite length
Symbol	Line AB or $\overleftrightarrow{AB}$	Line segment AB or $\overline{AB}$	Ray AB or $\overrightarrow{AB}$

Teacher's Note

Teacher should help the students to build the concept of line segment and differentiate it with line and ray.

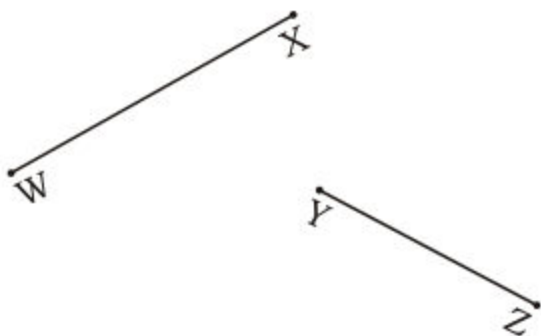
Add measures of two or more line segments

Adding measures of two or more line segments, means to find an another line segment whose length is the sum of measures of two given line segments.

Example: Add the given line segments

$$m\overline{WX} = 4 \text{ cm and } m\overline{YZ} = 3 \text{ cm.}$$

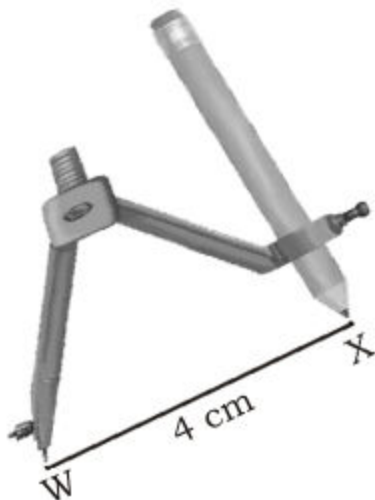
Using pair of compasses, pencil and ruler.



1. Draw a line m of any suitable length and mark a point A on it.



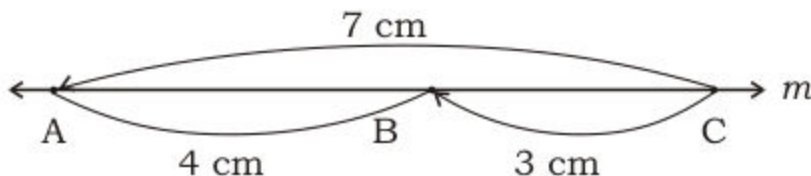
2. Use pair of compasses to measure the length of $\overline{WX} = 4 \text{ cm}$.



3. Keep the radius equals to 4 cm, with centre at point A, draw an arc to cut line m at point B.



4. Repeat step 2 to get radius equal to 3 cm and with centre B, draw an arc to cut line m in the same direction at point C.



Therefore, $m\overline{AC} = m\overline{AB} + m\overline{BC} = 4 \text{ cm} + 3 \text{ cm} = 7 \text{ cm}$

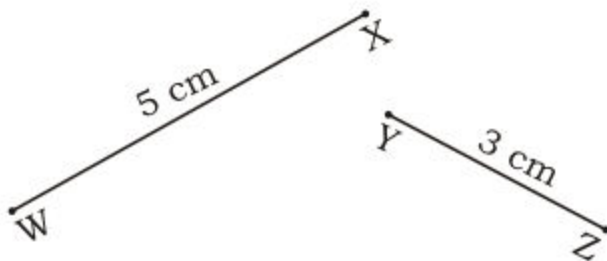
Hence $\overline{AC}$ is the required line segment which shows the sum of measures of two line segments.

We can also add measures of more than two line segments in the same way.

Subtract measure of line segment from a longer one

For subtracting measure of one line segment from a longer one, we are going to find another line segment whose measure is the difference of measures of two given line segments.

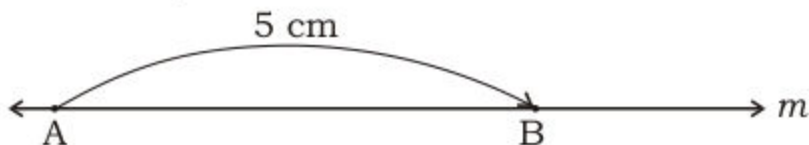
Example: Subtract the measure of line segment $\overline{YZ}$ from the measure of line segment $\overline{WX}$. Using pair of compasses, pencil and ruler.



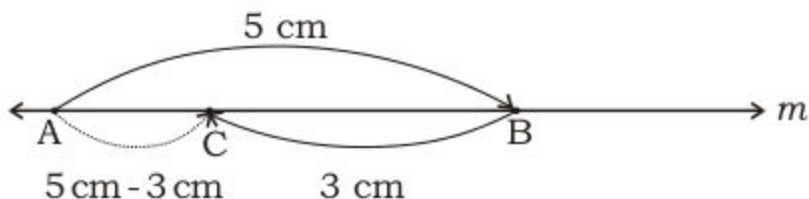
1. Draw a line m of any suitable length and mark a point A on it.



2. Measure the length of $\overline{WX}$ with centre A, draw an arc to cut line m at point B.



3. With centre B and radius equals to $\overline{YZ}$ draw another arc to cut line m at point C in the opposite direction.



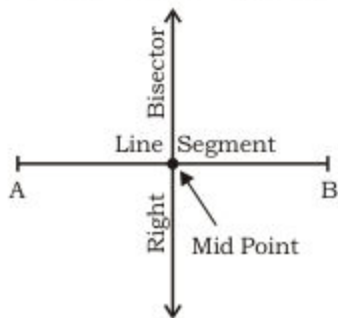
Therefore, $m\overline{AC} = m\overline{WX} - m\overline{YZ} = 5 \text{ cm} - 3 \text{ cm} = 2 \text{ cm}$

Hence $\overline{AC}$ is the required line segment which shows the difference of a line segment from a longer one.

Draw a right bisector of a given line segment using compasses

Right bisector of a given line segment is a line which makes an angle of 90° with the given line segment and divide it into two equal parts.

In the given figure the line ℓ is the right bisector of $\overline{AB}$.



Note: Right bisector is also known as perpendicular bisector.

Example: Draw a right bisector of $\overline{WX}$ measuring 6 cm by using pair of compasses.

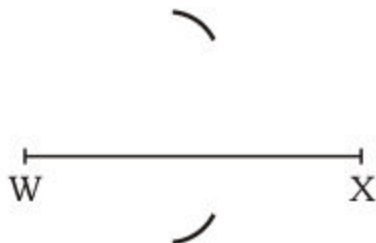
Consider the given line segment WX measuring 6 cm.



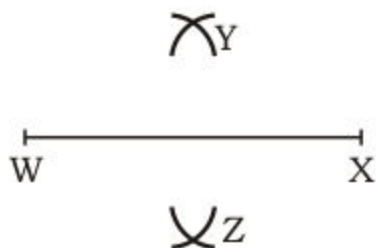
Now follow the following steps by using pair of compasses, pencil and ruler.

Steps of construction:

Step I: With centre W and radius more than half of $\overline{WX}$ (say 3.5 cm), draw two arcs on each side of $\overline{WX}$.



Step II: With centre X draw two arcs of same radius to cut each other at points Y and Z .



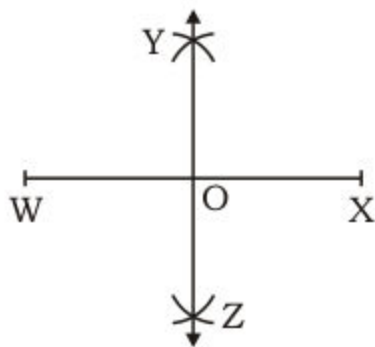
Step III: Draw $\overleftrightarrow{YZ}$ which cuts $\overline{WX}$ at point O .

Therefore $\overleftrightarrow{YZ}$ bisects the given line segment WX at point O .

as $m\overline{WO} = m\overline{OX}$.

Hence $\overleftrightarrow{YZ}$ is the required right bisector of $\overline{WX}$.

Note: $\overrightarrow{OY}$ and $\overrightarrow{OZ}$ are also right bisectors of $\overline{WX}$.



Draw a perpendicular to a given line from a point on it using compasses

Example: Draw a perpendicular to the line PQ at point R on it by using pair of compasses.

Steps of construction:

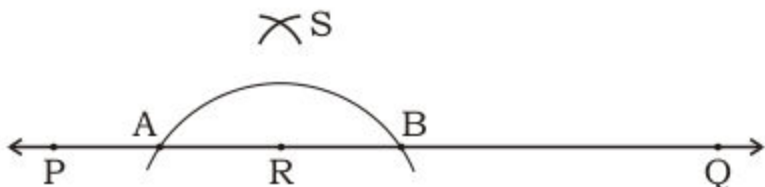
Step I: Draw a line PQ, mark point R on it.



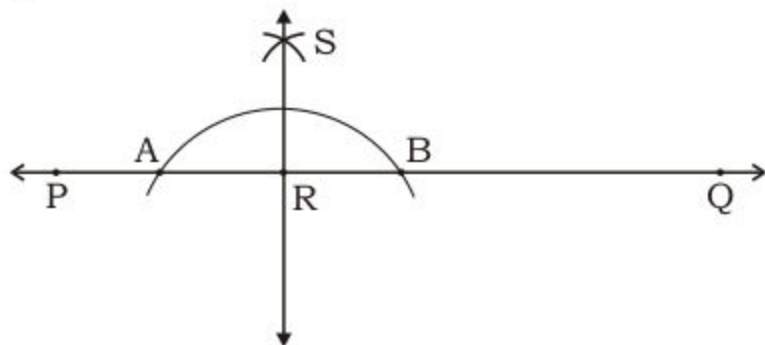
Step II: With centre R draw an arc of suitable radius to cut $\overleftrightarrow{PQ}$ at points A and B, such that $m\overline{RA} = m\overline{RB}$.



Step III: With centre A and B draw two arcs of radius more than $m\overline{RA}$, to cut each other at point S.



Step IV: Join point R and S, and extend it on both sides of line $\overleftrightarrow{PQ}$ to form $\overleftrightarrow{RS}$.



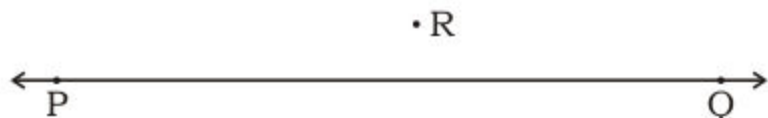
$\overleftrightarrow{RS}$ is the required perpendicular to $\overleftrightarrow{PQ}$ from a point R on it.

Draw a perpendicular to a given line, from a point outside the line, using compasses

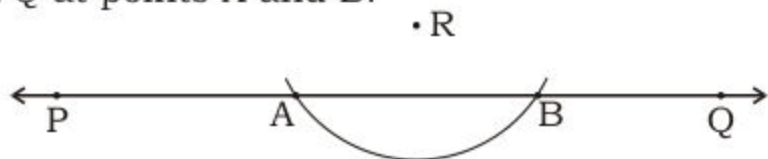
Example: Draw a perpendicular to the line PQ from a point R outside the line by using pair of compasses.

Steps of construction:

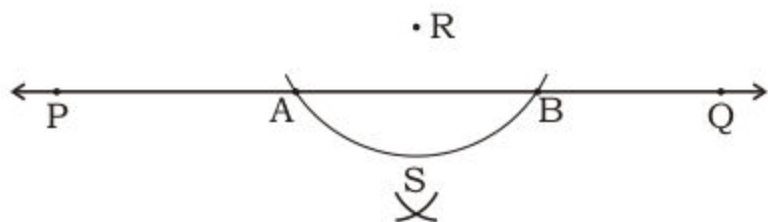
Step I: Draw a given line PQ, mark a point R outside it.



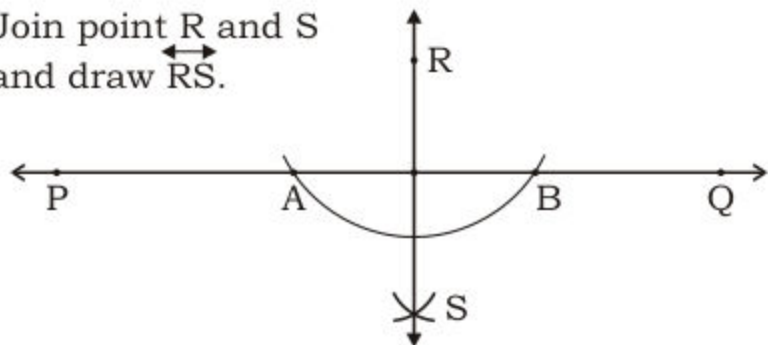
Step II: With centre R draw an arc of suitable radius to cut PQ at points A and B.



Step III: With centres A and B, draw two arcs of radius more than half of $\overline{AB}$, to cut each other at point S (in the opposite direction of R).



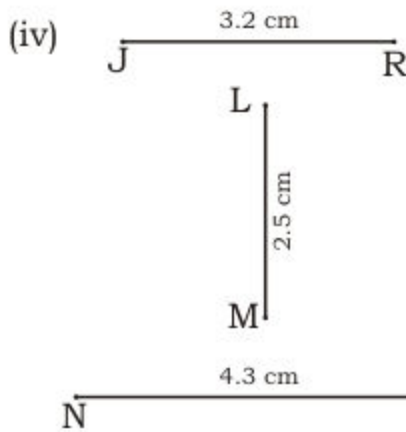
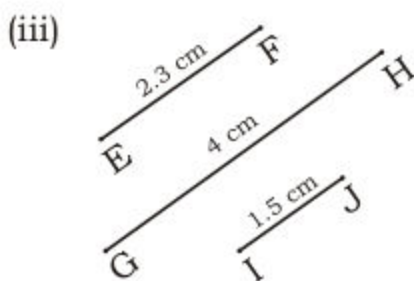
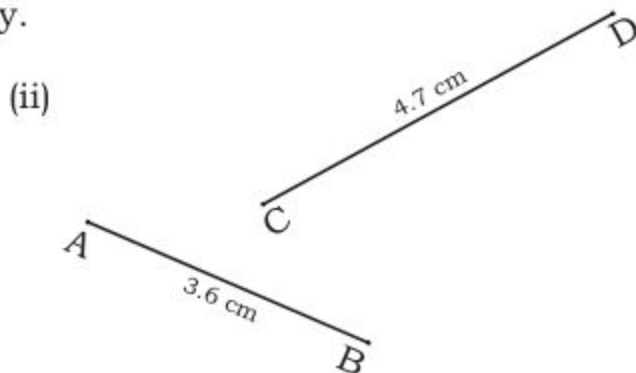
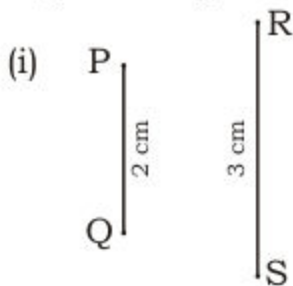
Step IV: Join point R and S and draw $\overleftrightarrow{RS}$.



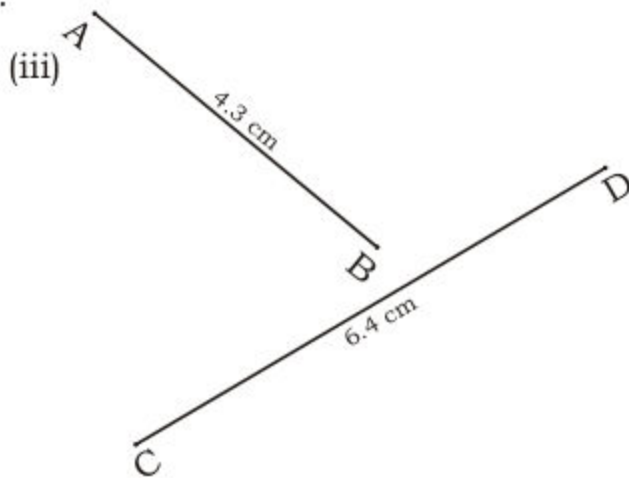
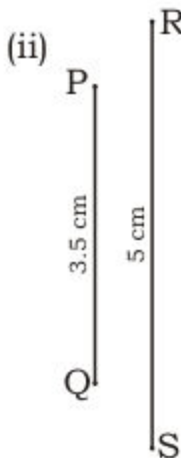
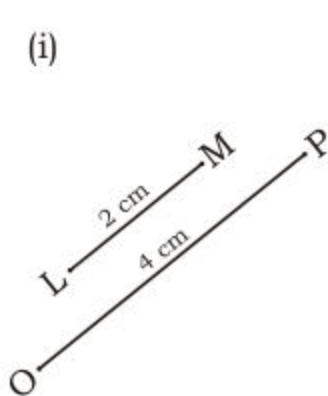
$\overleftrightarrow{RS}$ is the required perpendicular of PQ from a point R outside it.

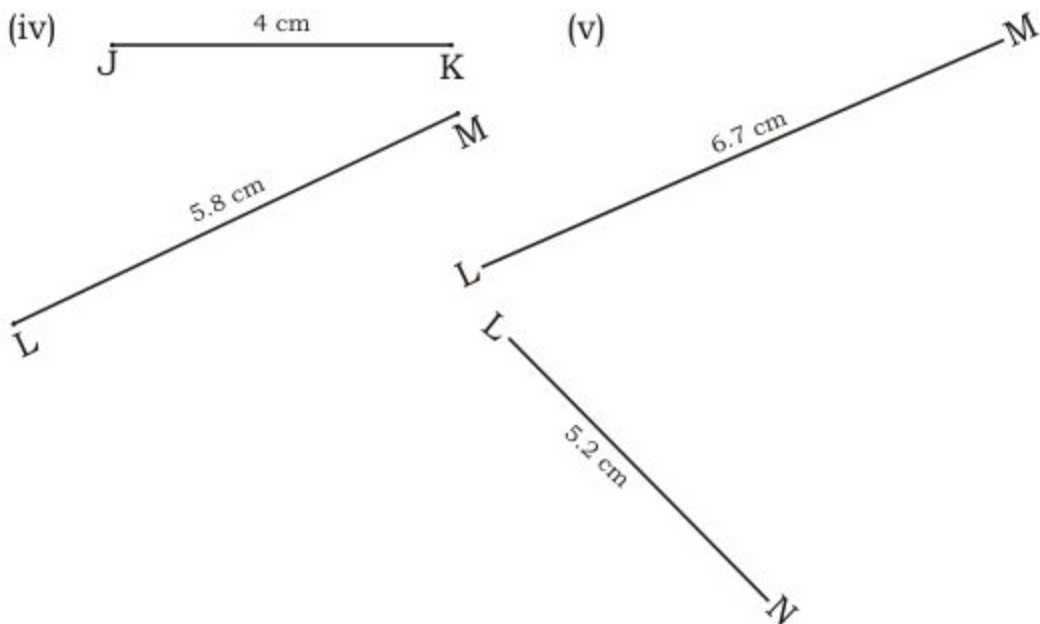
EXERCISE 10.1

1. Find the sum of measures of the following pairs of line segments geometrically.



2. Find the difference of measures of the following line segments geometrically.

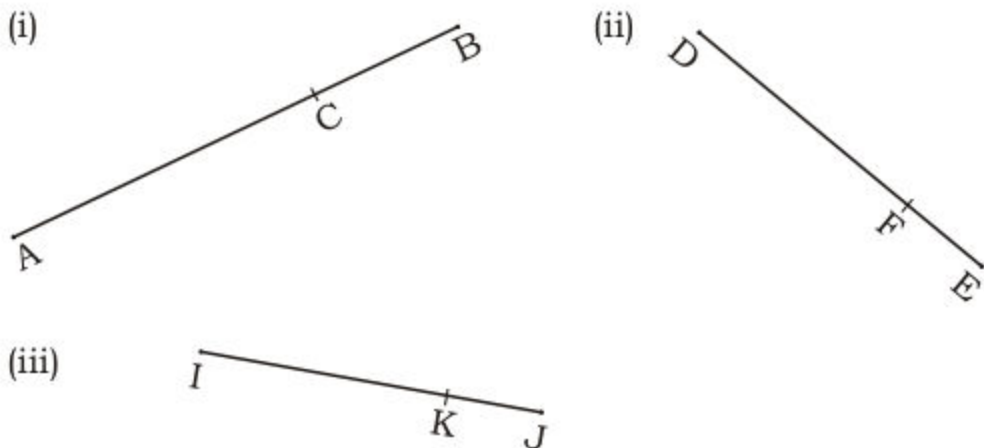




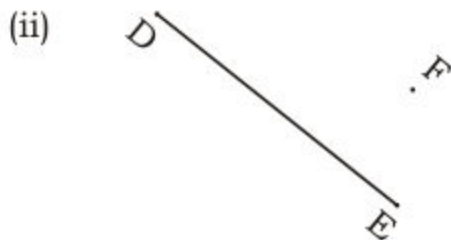
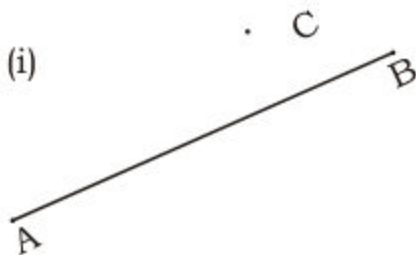
3. Draw right bisectors of the line segments of the following measures.

- (i) 5 cm (ii) 6.3 cm (iii) 8 cm

4. Draw perpendicular from given points on the following line segments.



5. Draw perpendiculars from given points outside the following line segments.

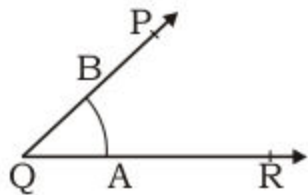


10.2 CONSTRUCTION OF ANGLES

We already know how to construct angles of various measures by using protractor. Let us learn the same with the help of compasses through examples.

Use compasses to construct an angle equal in measure of a given angle

Example: Construct an angle equal in measure of the angle PQR.



Consider the given angle PQR.

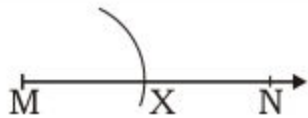
Steps of construction:

Step I: Draw $\overrightarrow{MN}$

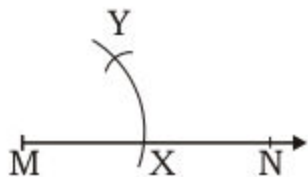


Step II: With center Q draw an arc of any suitable radius, to cut $\overrightarrow{QR}$ and $\overrightarrow{QP}$ at points A and B respectively.

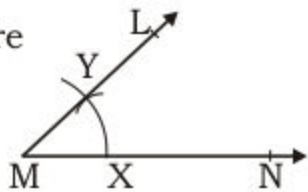
Step III: With centre M, draw an arc of radius equals to $m\overline{QA}$ to cut $\overrightarrow{MN}$ at point X.



Step IV: With centre X and taking same radius draw an arc to cut the previous arc at point Y.



Step V: Join point M with Y and extend to the point L through point Y. Measure it with protractor as $m\angle LMN = m\angle PQR$

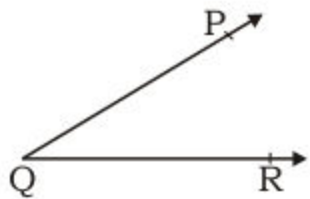


Hence $\angle LMN$ is the required angle.

Use compasses to construct an angle twice in measure of a given angle

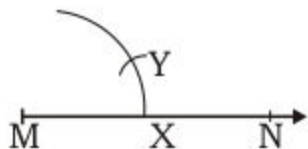
Example: Construct an angle twice in measure of the given angle PQR.

Consider the given angle PQR and measure it.

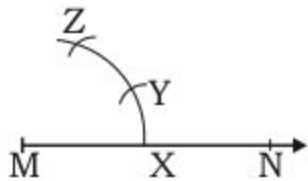


Steps of construction:

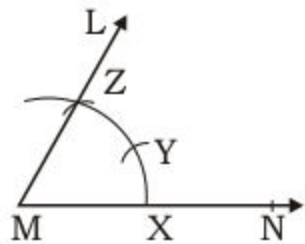
Step I: Now follow the steps from I to IV of the previous example and get the arc XY as shown in figure.



Step II: Taking same radius with centre Y, draw another arc to cut the previous arc at point Z.



Step III: Draw $\overrightarrow{ML}$ passing through point Z.



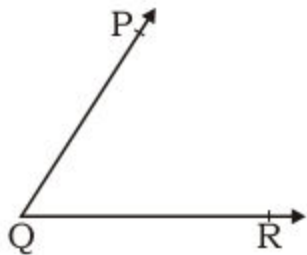
Here $\angle LMN$ is the required angle as $m\angle LMN = 2 m\angle PQR$.

We can verify measure with the help of protractor as the required angle is double of angle $\angle PQR$

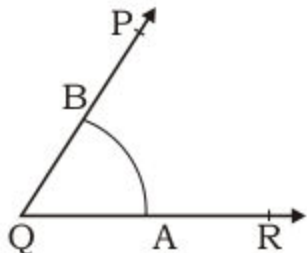
Use compasses to bisect a given angle

Example: Draw the bisector of the angle PQR.

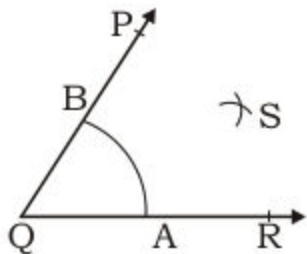
Consider the given angle PQR.


Steps of construction:

Step I: With centre Q draw an arc of any suitable radius to cut $\overrightarrow{QR}$ and $\overrightarrow{QP}$ at points A and B respectively.

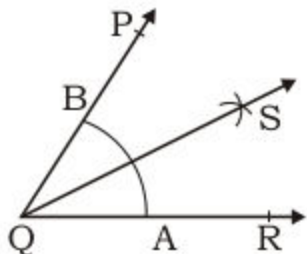


Step II: With centres A and B draw two arcs of suitable radius to cut each other at point S.



Step III: Join point Q with S and draw $\overrightarrow{QS}$.

Here $\overrightarrow{QS}$ bisects $\angle PQR$ in two equal angles i.e. $m\angle PQS = m\angle RQS$



We can verify the result by measuring both the angles with protractor.

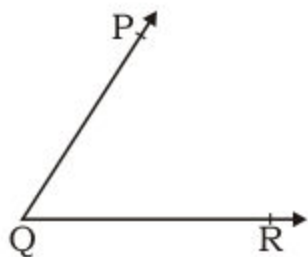
Use compasses to divide a given angle into four equal angles

Example: Divide the angle PQR into four equal angles.

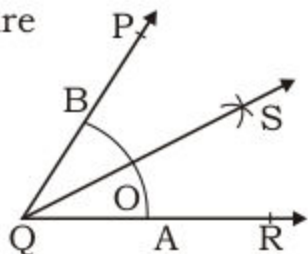
Consider the given angle PQR.

Steps of construction:

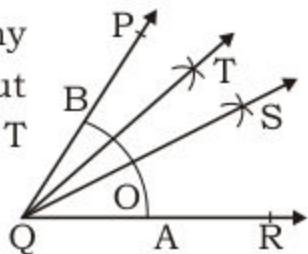
Step I: First of all bisect the given angle PQR in two equal angles PQS and RQS using compass as shown in the figure



Step II: Name the intersecting point of $\overrightarrow{QS}$ as O.



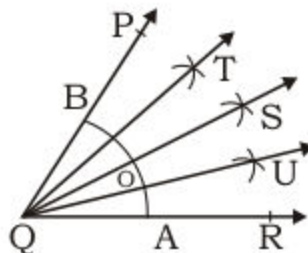
Step III: With centres O and B and taking any suitable radius, draw two arcs to cut each other at point T. Join Q with T to get $\overrightarrow{QT}$.



Step IV: With centres O and A, repeat the same process to get $\overrightarrow{QU}$.

Now the given $\angle PQR$ has been divided into four equal angles.

i.e. $m\angle PQT = m\angle TQS = m\angle SQU = m\angle UQR$



Note: We can verify measure with the help of protractor.

Use compasses to construct the following angles:

60° , 30° , 15° , 90° , 45° , $(22\frac{1}{2})^\circ$, 75° , $(67\frac{1}{2})^\circ$, 120° , 150° , 165° , 135° , 105°

1: Construction of angle of 60° :

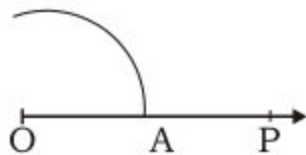
Angle of 60° is the basic angle in construction of angles by using pair of compasses.

Steps of construction:

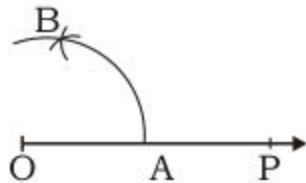
Step I: Draw $\overrightarrow{OP}$.



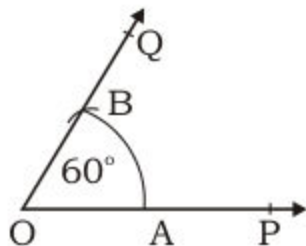
Step II: With centre O draw an arc of any suitable radius to cut $\overrightarrow{OP}$ at point A.



Step III: With centre A, draw another arc of the same radius to cut the previous arc at point B.



Step IV: Join points O and B and extend to the point Q.



Hence $\angle POQ$ is the required angle of 60° ,
i.e. $m\angle POQ = 60^\circ$.

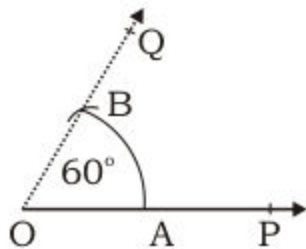
Note: We can verify it by using protractor.

2: Construction of angle of 30° :

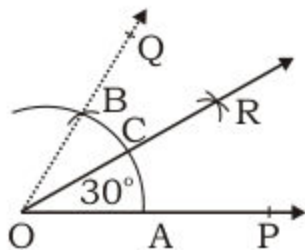
As $\frac{60^\circ}{2} = 30^\circ$, so we construct an angle of 60° and bisect it in two equal angles of 30° .

Steps of construction:

Step I: First construct an angle of 60° by using compass as shown in the figure.



Step II: With centres A and B draw two arcs of any suitable radius to cut each other at point R.



Step III: Join O and R and to draw $\overrightarrow{OR}$ which cuts the arc AB at point C.

Here, $m\angle POR = 30^\circ$ is the required angle.

We can measure the angles $\angle POR$ and $\angle ROQ$ by using protractor.

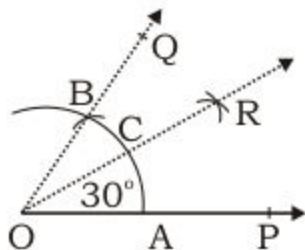
Note: As $m\angle QOP$ is bisected, so $m\angle QOR$ is also of 30° .

3: Construction of angle of 15° :

As $(\frac{30}{2})^\circ = 15^\circ$, so we construct an angle of 30° and bisect it to get the angle of 15° .

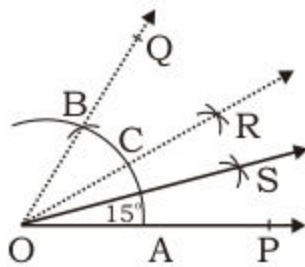
Steps of construction:

Step I: First construct angle of 30° as shown in the figure.



Step II: Name the intersecting point of $\overrightarrow{OR}$ as C.

Step III: With centres C and A and draw two arcs of any convenient radius to cut each other at point S.



Step IV: Join O to S to get $\overrightarrow{OS}$

Here $m\angle SOP = 15^\circ$ is the required angle.

Measure the angles SOP and ROS by using protractor.

Note: As $\angle ROP$ is bisected, so $m\angle ROS$ is also of 15° .

4: Construction of angle of 90° :

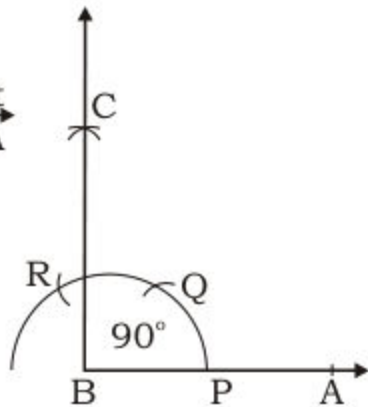
As $60^\circ + 30^\circ = 90^\circ$, so we construct an angle of 60° twice and then bisect its outer angle to get the angle of 90° .

Steps of construction:

Step I: Draw $\overrightarrow{BA}$.

Step II: With centre B and any convenient radius, draw an arc to cut $\overrightarrow{BA}$ at point P.

Step III: Keeping the radius same with centre P, draw another arc to cut the previous arc at point Q. (It is the mark for our first 60° angle).



Step IV: With centre Q and same radius, draw another arc to cut the previous longer arc at point R (It is the mark for 2^{nd} angle of 60°).

Step V: With centres Q and R and same radius, draw two arcs to cut each other at point C (Bisecting the 2^{nd} angle of second 60° to get measure of 30°).

Step VI: Join points B and C and extend to form $\overrightarrow{BC}$.

Here $m\angle ABC = 90^\circ$ is the required angle.

Measure the angle ABC by using protractor.

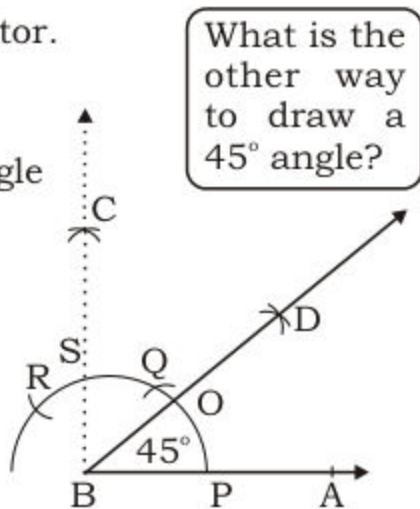
5: Construction of angle of 45° :

As $(\frac{90}{2})^\circ = 45^\circ$, so we will construct an angle of 90° and bisect it to get the angle of 45° .

Steps of construction:

Step I: First construct angle of 90° as shown in the figure.

Step II: With the centre B draw an arc PS to cut $\overrightarrow{BA}$ at P and $\overrightarrow{BC}$ at S.



What is the other way to draw a 45° angle?

Step 3: With centres S and P and any convenient or similar radii, draw two arcs to cut each other at point D.

Step 4: Join point B to D and extend it to get $\overrightarrow{BD}$.

Measure the angle ABD by using protractor as measurement is 45° .

$m\angle ABD = 45^\circ$. It is the required angle.

Note: As $\angle ABC$ is bisected, so $\angle CBD$ is also of 45° .

6: Construction of angle of $(22\frac{1}{2})^\circ$:

As $(\frac{45}{2})^\circ = (22\frac{1}{2})^\circ$, so we construct an angle of 45° .

and bisect it equally to get the angle of $(22\frac{1}{2})^\circ$.

Steps of construction:

First construct angle of 45° and follow the following steps:

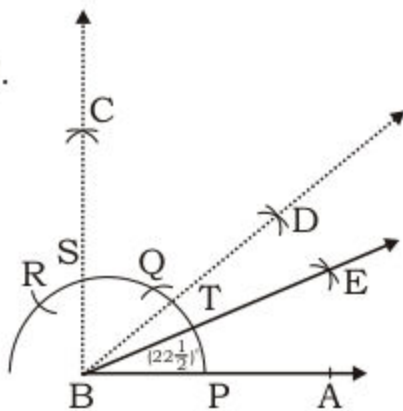
Step I: Name the intersecting point of $\overrightarrow{BD}$ as T.

Step II: With centres T and P and any convenient similar radii, draw two arcs to cut each other at point E.

Step III: Join B to E and extend it to get $\overrightarrow{BE}$.

Hence $m\angle ABE = (22\frac{1}{2})^\circ$ is the required angle.

Note: As $\angle ABD$ is bisected, so, $\angle DBE$ is also of $(22\frac{1}{2})^\circ$.



7: Construction of angle of 75° :

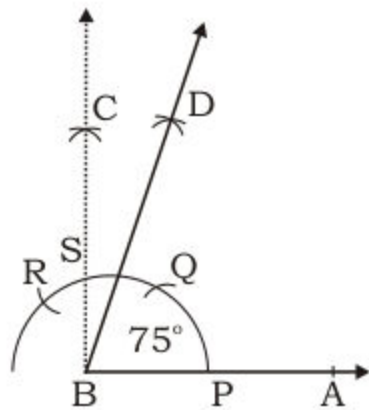
As $90^\circ - 15^\circ = 75^\circ$, so we will first construct a 90° angle then subtract 15° from it by bisecting the angle of 30° between 60° and 90° angles to get the angle of 75° .

Steps of construction:

Step I: First construct an angle of 90° as shown in the figure.

Step II: Name the intersecting point of arc PR with $\overrightarrow{BC}$ as S.

Step III: With centres S and Q draw two arcs of equal radius to cut each other at point D.



Step IV: Join point B to D and extend it to get $\overrightarrow{BD}$.

Measure the angle ABD by using protractor as $m\angle ABD = 75^\circ$. Hence it is the required angle.

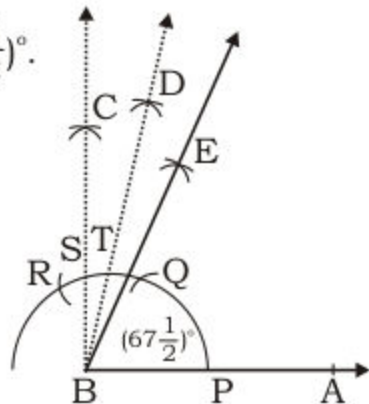
8: Construction of angle of $(67\frac{1}{2})^\circ$:

As $75^\circ - (7\frac{1}{2})^\circ = (67\frac{1}{2})^\circ$, so we will first construct an angle of 75° and then subtract $(7\frac{1}{2})^\circ$ from it by bisecting the angle of 15° between 60° and 75° angles to get the angle of $(67\frac{1}{2})^\circ$.

Steps of construction:

Step I: First construct an angle of 75° as shown in the figure.

Step II: Name the intersecting point of arc PR with $\overrightarrow{BD}$ as T.



Step III: With centres T and Q draw two arcs of any same radius, to cut each other at point E. (Which gives us $(7\frac{1}{2})^\circ$ measure after 60°)

Step IV: Join point B with E and extend it to get $\overrightarrow{BE}$.

Measure the angle ABE by using protractor $m\angle ABE = (67\frac{1}{2})^\circ$. Hence it is the required angle.

9: Construction of angle of 120° :

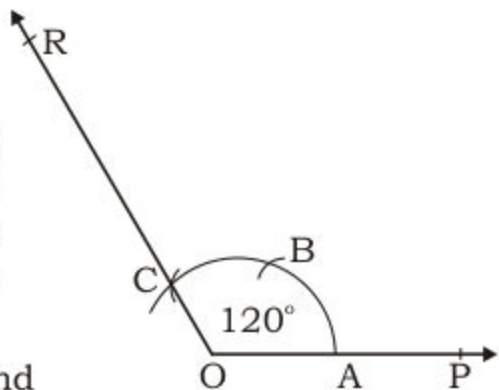
As $60^\circ + 60^\circ = 120^\circ$, so we will construct 60° angle twice to get the angle of 120° .

Steps of construction:

Step I: Follow the process of constructing angle of 60° .

Step II: With centre B and keeping the radius same, draw another arc to cut the previous large arc AB at point C.

Step III: Join point O to C and extend it to form $\overrightarrow{OR}$.



Measure the angle POR by using protractor $m\angle POR = 120^\circ$. Hence it is the required angle.

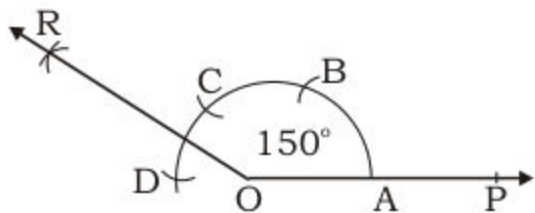
10: Construction of angle of 150° :

As $180^\circ - 30^\circ = 150^\circ$, so we will construct an angle of 60° thrice to get an angle of 180° and then subtract 30° from it by bisecting the last portion of 60° between 120° and 180° to get the angle of 150° .

Steps of construction:

Step I: First construct an angle of 120° .

Step II: With centre C and keeping the radius same, draw another arc to cut the arc AC thrice at point D. (It will give us the angle of 180°).



Step III: With centres C and D and some convenient similar radius, draw two arcs to cut each other at point R.

Step IV: Join point O to R and extend it to form $\overrightarrow{OR}$.

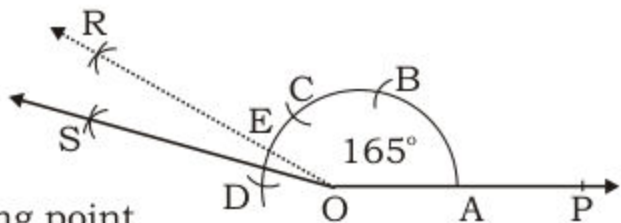
Measure the angle POR by using protractor $m\angle POR = 150^\circ$. Hence it is the required angle.

11: Construction of angle of 165° :

As $150^\circ + 15^\circ = 165^\circ$, so we will first construct an angle of 150° and bisect 30° of its outer part to get the angle of 165° .

Steps of construction:

First construct an angle AOR of 150° as shown in the figure and follow the following steps:



Step I: Name the intersecting point of the arc AD with $\overrightarrow{OR}$ as E.

Step II: With centres E and D draw two arcs of any convenient same radii, to cut each other at point S.

Step III: Join point O with S and extend it to get $\overrightarrow{OS}$.

Measure the angle POS by using protractor $m\angle POS = 165^\circ$. Hence it is the required angle.

12: Construction of angle of 135° :

As $90^\circ + 45^\circ = 135^\circ$, so we will construct an angle of 90° and then an angle of 45° by bisecting its outer part to get the angle of 135° .

Steps of construction:

First construct an angle of 90° as shown in the figure and follow the following steps:

Step I: Name the intersecting point of arc PR with $\overrightarrow{BC}$ as T.

Step II: With centre R and keeping the radius of arc QR, draw another arc to cut the base arc thrice at point S.

Step III: With centre S and T and any suitable radius, draw two arcs to cut each other at point D.

Step IV: Join point B with D to get $\overrightarrow{BD}$.

Measure the angle ABD by using protractor $\angle ABD = 135^\circ$. Hence it is the required angle.

13: Construction of angle of 105° .

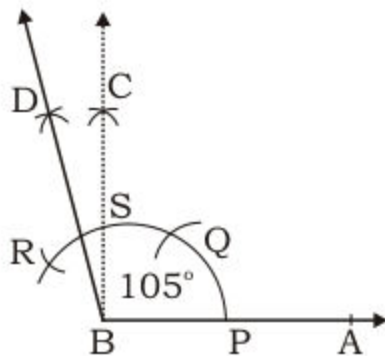
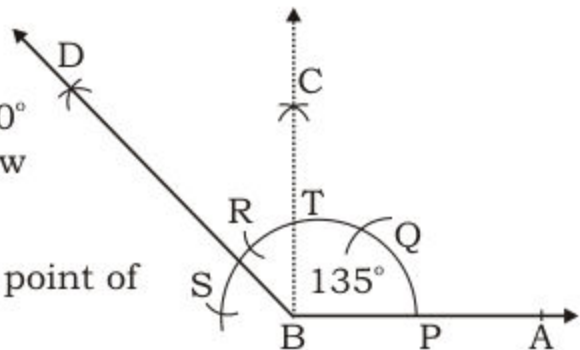
As $90^\circ + 15^\circ = 105^\circ$ so, we will construct an angle of 90° and then bisect its outer portion of 30° between 90° and 120° to get the angle of 105° .

Steps of construction:

First construct an angle of 90° as shown in the figure and follow the following steps:

Step I: Name the intersecting point of arc PR with $\overrightarrow{BC}$ as S.

Step II: With centres R and S and any convenient same radii, draw two arcs to cut each other at point D.



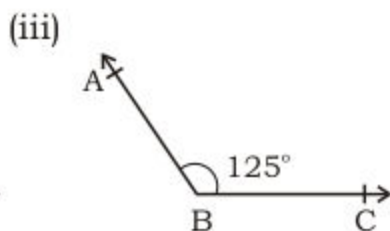
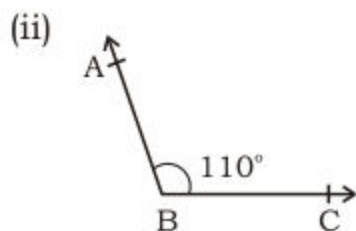
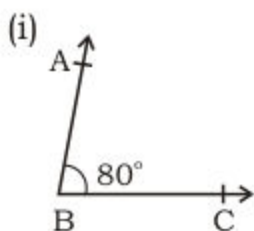
Step III: Join point B with D to get $\overrightarrow{BD}$.

Measure the angle ABD by using protractor $m\angle ABD = 105^\circ$.

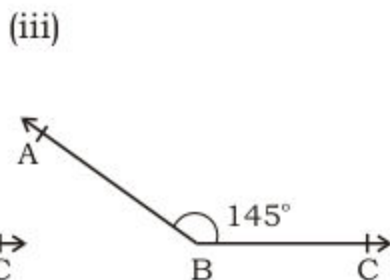
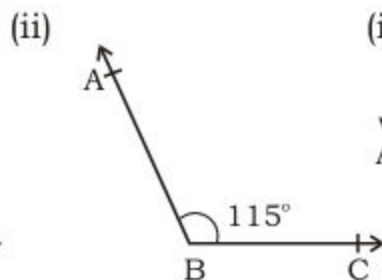
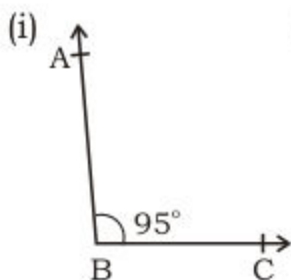
Hence it is the required angle.

EXERCISE 10.2

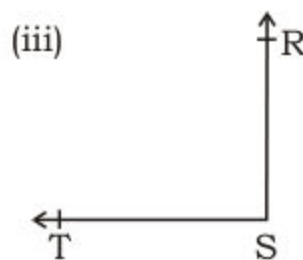
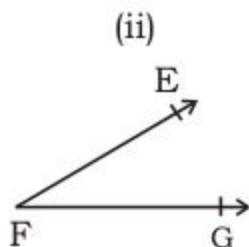
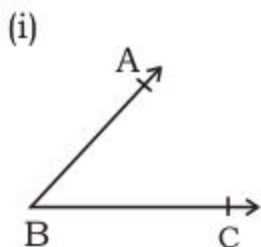
1. Bisect the following angles with the help of compasses.



2. Divide the following angles with the help of compasses into four equal angles.



3. Draw the angle of equal measures and twice of the following angles with the help of pair of compass and ruler. Check your answers with protractor.



4. Draw the following angles with the help of pair of compasses and ruler.

- (i) 30° (ii) $(22\frac{1}{2})^\circ$ (iii) $(67\frac{1}{2})^\circ$ (iv) 75°
 (v) 105° (vi) 120° (vii) 135° (viii) 165°

10.3 CONSTRUCTION OF TRIANGLES

Triangle is a plane closed geometrical figure bounded by three sides. It has three sides and three angles.

We are going to learn different ways of constructing triangles according to the given elements.

Construct a triangle when three sides (SSS) are given.

Caution: Sum of two sides should be greater than the third side

Example: Construct a triangle ABC when

$m\overline{AB} = 4 \text{ cm}$, $m\overline{BC} = 6 \text{ cm}$ and $m\overline{AC} = 7 \text{ cm}$.

Steps of construction:

Step I: Draw $m\overline{AC} = 7 \text{ cm}$.

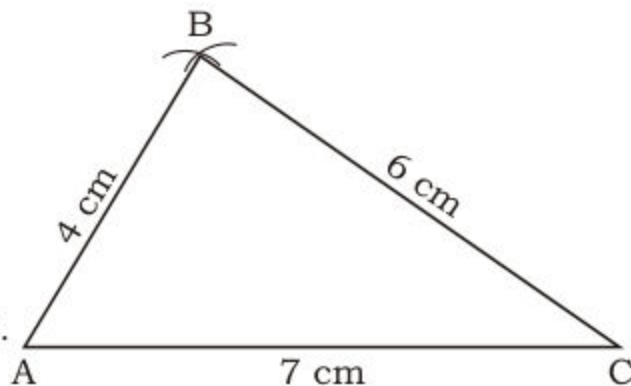
Step II: With centre A and radius equal to 4 cm, draw an arc.

Step III: With centre C and radius equal to 6 cm, draw another arc to cut the previous arc at point B.

Step IV: Join point B to A and C.

The $\triangle ABC$ is constructed.

Can we construct a triangle with sides measures 4 cm, 3 cm and 2 cm?



Construction of triangle is possible only if the sum of any two sides is greater than the third side.

Construct a triangle when two sides and their included angle (SAS) are given

Example: Construct a triangle DEF when $m\overline{DE} = 6$ cm, $m\overline{DF} = 4$ cm and $m\angle D = 70^\circ$.

Steps of construction:

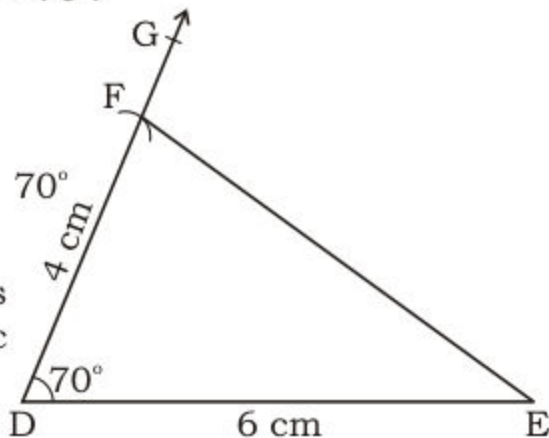
Step I: Draw $m\overline{DE} = 6$ cm.

Step II: At point D draw $m\angle EDG = 70^\circ$ by using protractor.

Step III: With centre D and radius equal to 4 cm, draw an arc to cut $\overrightarrow{DG}$ at point F.

Step IV: Join point F with E.

Hence, the required $\triangle DEF$ is constructed.



Construct a triangle when two angles and included side (ASA) are given

Example: Construct a triangle MNO when $m\overline{NO} = 5$ cm, $m\angle N = 60^\circ$ and $m\angle O = 55^\circ$.

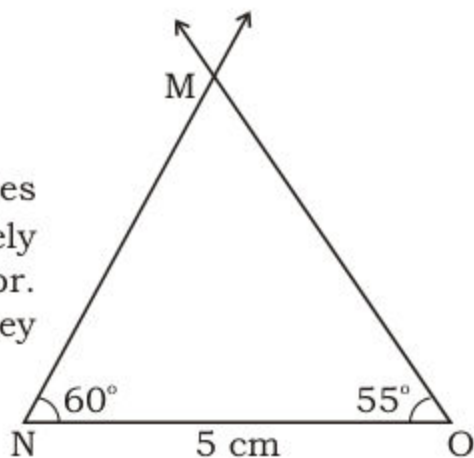
Steps of construction:

Step I: Draw $m\overline{NO} = 5$ cm.

Step II: At points N and O, draw angles of 60° and 55° respectively with the help of protractor. Name the point where they intersect each other.

Step III: The arms of both angles intersect at point M.

Hence, the required $\triangle MNO$ is constructed.



Construct a triangle when hypotenuse and one side (RHS) for a right angled triangle are given

Example: Construct a right angled triangle PQR when hypotenuse $\overline{PQ}$ is 7 cm long and a side $\overline{QR}$ is 5 cm long and $m\angle R = 90^\circ$.

Steps of construction:

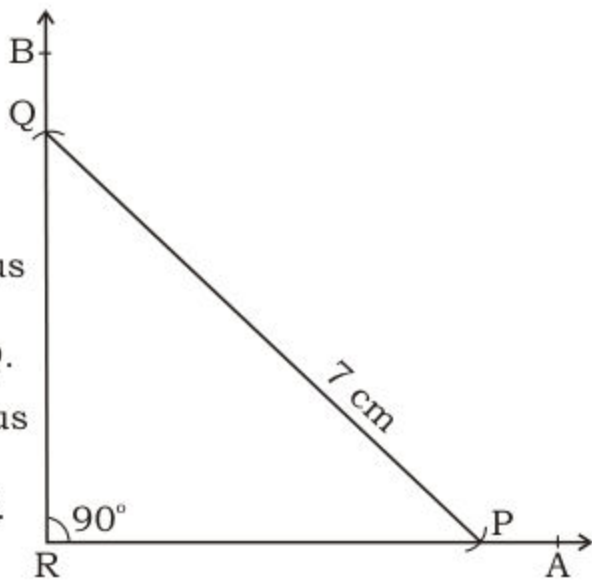
Step I: Draw $\overrightarrow{RA}$ greater than 7 cm.

Step II: At point R draw $m\angle ARB = 90^\circ$.

Step III: With centre R and radius equal to 5 cm, draw an arc to cut $\overrightarrow{RB}$ at point Q.

Step IV: With centre Q and radius equal to 7 cm, draw an arc to cut $\overrightarrow{RA}$ at point P.

Step V: Join point P and Q.



Hence, the required ΔPQR is constructed and it is the right angled triangle.

EXERCISE 10.3

Construct a triangle ABC when:

1. $m\overline{AB} = 3$ cm, $m\overline{BC} = 5$ cm and $m\overline{CA} = 7$ cm.
2. $m\overline{AB} = 2$ cm, $m\overline{BC} = 5$ cm and $m\overline{CA} = 5$ cm.
3. $m\overline{AB} = 5$ cm, $m\overline{BC} = 7$ cm and $m\overline{CA} = 9$ cm.
4. $m\overline{AB} = 8$ cm, $m\overline{BC} = 6$ cm and $m\overline{CA} = 10$ cm.

Teacher's Note

Teacher should help the students to construct the triangles of different measures on blackboard as well as in their copies by geometrical instruments.

Construct a triangle MNO when:

5. $m\angle N = 63^\circ$, $m\overline{NO} = 3$ cm and $m\overline{MN} = 5$ cm.
6. $m\angle N = 75^\circ$, $m\overline{NO} = 3.4$ cm and $m\overline{MN} = 4$ cm.
7. $m\angle N = 80^\circ$, $m\overline{NO} = 3.5$ cm and $m\overline{MN} = 4.3$ cm.
8. $m\angle N = 90^\circ$, $m\overline{NO} = 4$ cm and $m\overline{MN} = 3$ cm.

Construct a triangle PQR when:

9. $m\overline{PQ} = 5$ cm, $m\angle Q = 40^\circ$, $m\angle P = 60^\circ$.
10. $m\overline{PQ} = 7$ cm, $m\angle P = 65^\circ$, $m\angle Q = 60^\circ$.
11. $m\overline{QR} = 6.5$ cm, $m\angle Q = 45^\circ$, $m\angle R = 30^\circ$.
12. $m\overline{QR} = 7.5$ cm, $m\angle Q = 35^\circ$, $m\angle R = 40^\circ$.

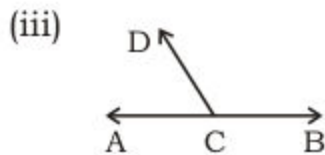
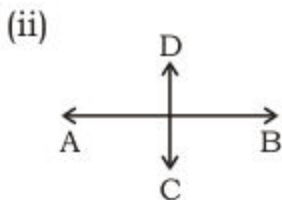
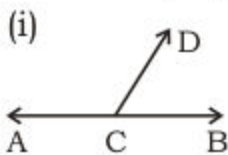
Construct a right angled triangle XYZ when $m\angle Z = 90^\circ$.

13. $m\overline{XY} = 7$ cm, $m\overline{XZ} = 5$ cm.
14. $m\overline{XY} = 10$ cm, $m\overline{XZ} = 6$ cm.
15. $m\overline{XY} = 14$ cm, $m\overline{XZ} = 10$ cm.
16. $m\overline{XY} = 8$ cm, $m\overline{XZ} = 6.5$ cm.

REVIEW EXERCISE 10

1. Give four examples of objects which represent the line segments from your environment.
2. If $m\overline{AB} = 4.3$ cm, $m\overline{CD} = 3.5$ cm and $m\overline{EF} = 2$ cm, construct geometrically a line segment whose length is equal to:
 - (i) $m\overline{AB} + m\overline{CD} + m\overline{EF}$
 - (ii) $m\overline{AB} - m\overline{CD}$
 - (iii) $m\overline{AB} - m\overline{EF}$

3. Draw a line segment measuring 7.6 cm and draw its right bisector. Then bisect its each part into equal parts.
4. Draw angles of the following measures with the help of compasses and bisect them.
 - (i) 30° (ii) 75° (iii) 105°
5. Construct a triangle ABC according to the following conditions and write steps of construction in each case.
 - (i) $m\overline{AB} = 8 \text{ cm}$, $m\overline{AC} = 6 \text{ cm}$, $m\angle A = 65^\circ$
 - (ii) $m\angle A = 45^\circ$, $m\angle C = 45^\circ$, $m\overline{AC} = 7.5 \text{ cm}$
 - (iii) $m\overline{AB} = 5 \text{ cm}$, $m\overline{AC} = 7 \text{ cm}$, $m\angle B = 90^\circ$
6. Draw is a line segment and name it.
7. Tick the perpendicular bisector of a line AB.



8. Choose the correct answer.

- (i) Line segment AB is denoted by:
 - (a) $\overleftrightarrow{AB}$ (b) $\overline{AB}$ (c) $\overrightarrow{AB}$ (d) $\overrightarrow{BA}$
- (ii) Bisector divides a line segment into _____ equal parts.
 - (a) two (b) three (c) four (d) five
- (iii) In $\overline{PQ}$, the right bisector passes through its:
 - (a) end point (b) mid point
 - (c) point P (d) point Q
- (iv) An angle is measured by:
 - (a) Ruler (b) Compasses
 - (c) Protractor (d) Set squares

9. Write True (T) or False (F) for the following:

- (i) A line has a definite length.
- (ii) A ray has two end points.
- (iii) An angle of 15° is formed by bisecting the angle of 30° .
- (iv) A triangle has 6 elements.
- (v) A triangle is formed when sum of measures of any two sides is less than the third side.

SUMMARY

- Line segment is a part of line. It has two distinct end points.
- Right bisector is a line, line segment or a ray, which is perpendicular to a given line and also bisects it.
- Line segments and angles can be bisected with the help of ruler and a pair of compasses.
- Angles of measures 60° , 30° , 15° , 90° , 45° , $(22\frac{1}{2})^\circ$, 75° , $(67\frac{1}{2})^\circ$, 120° , 150° , 165° , 135° and 105° can be drawn with the help of ruler and pair of compasses.
- Triangle is a closed geometrical figure bounded by three sides. It has three angles.
- A triangle can be constructed if one of the following set of elements of triangle are known:
 - (i) Measures of three sides (SSS).
 - (ii) Measures of two sides and included angle (SAS).
 - (iii) Measures of two angles and included side (ASA).
 - (iv) Measures of hypotenuse and one side (RHS).
- To construct a triangle by SSS, the sum of measures of any two sides must be greater than the third side.